DF-PEM: Dual-Flow Prompt-Expert Mamba for Multimodal Remote Sensing Incremental Classification

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***Abstract*—Existing deep learning-based multimodal remote sensing imagery (MRSI) classification (MRSIC) models rely on fixed-category paradigms and struggle to adapt to novel categories, primarily due to the catastrophic forgetting of vintage knowledge, which substantially impedes their practical applications. To overcome this challenge, we propose a novel dual-flow prompt-expert Mamba (DF-PEM) network for MRSI incremental classification (IC), which incorporates dedicated parallel branches for spectral and spatial feature extraction, synergized with a cross-modal fusion module to integrate modality-specific representations. The primary objective of DF-PEM is to capture discriminative MRSI features while preserving knowledge of previously seen categories during the incremental learning process. Specifically, DF-PEM integrates the prompt-driven assistant-expert (PAE) mechanism within the Mamba architecture to dynamically activate category-specific prompt experts per input, enabling seamless adaptation to emerging categories while mitigating knowledge erosion of prior classes. Furthermore, a spherical Gaussian sift (SGS) strategy is developed in the initial stage, which models the complex feature distributions of historical categories and neutralizes noise and outlier effects, yielding robust representative prototypes. Finally, a prototype-aware regularization classifier (PRC) that adaptively balances prediction bias between incremental and old categories by learning regularization coefficients, which strategically penalize parameter updates that deviate the model from historical knowledge, further enhancing knowledge retention and generalization. Extensive experiments on benchmark datasets demonstrate that the proposed DF-PEM significantly outperforms existing methods, validating its effectiveness for MRSI classification tasks.**

***Index Terms*—Incremental learning, Mamba, multimodal remote sensing imagery classification (MRSIC), prompt.**

I. Introduction

MULTIMODAL remote sensing imagery (MRSI) significantly enhances the precise discrimination of land cover pixels by integrating complementary features derived from diverse data sources [e.g., multispectral imagery (MSI), hyperspectral imagery (HSI), and light detection and ranging (LiDAR)]. Consequently, MRSI classification (MRSIC) plays a pivotal role in Earth observation missions, supporting diverse applications such as industrial production [1], agricultural monitoring [2], and environmental monitoring [3], to name a few.

Over the past few decades, deep learning-based MRSIC networks have advanced rapidly, driven by their robust feature extraction capabilities. With the booming advancement of convolutional neural networks (CNNs), the CNN-based MRSI classification networks [4], [5], [6], [7], [8], [9], [10], [11] have exhibited exceptional classification results. For instance, Xu et al. [4] have designed a two-tunnel CNN framework to extract spatial–spectral features from HSI and a cascade network to capture LiDAR features and then fused multimodal features to enhance classification performance. Subsequent innovation on the dual-branch network, the representation-enhanced status replay network (RSRNet) [12], is developed to further enhance feature representation through dual augmentation (modal and semantic) to reduce representation bias, alleviating classifier bias with a status replay strategy to stabilize the decision boundary.

Nevertheless, the above CNN-based methods face inherent deficiencies in capturing long-range dependencies and global interactions between features, hindering further classification performance improvements. To address these limitations, several Transformer-based MRSIC networks [13], [14], [15], [16], [17] emerged, leveraging self-attention mechanisms to extract global relationships on multimodal data. In particular, this incorporates a dual-branch architecture consisting of a hierarchical CNN and Transformer (HCT) [18], thus leveraging the Transformer's robust capacity for modeling long-range dependencies and capturing intricate global relationships within the MRSI. To boost the information interaction across different modalities, Roy et al. [17] extend the traditional self-attention by introducing cross-hyperspectral and LiDAR (Cross-HL). Furthermore, a global clue-guided cross-memory quaternion Transformer network (GCCQTNet) [15] is proposed to overcome the impact of heterogeneity, which updates the local and global representations by considering the global information as an agent to suppress the negative impact of multimodal heterogeneity layer by layer. Moreover, the multimodal informative Vision Transformer (MIViT) [16] is designed, incorporating oriented attention fusion (OAF) to reduce redundant infor-

Received 12 December 2025; revised 22 January 2026; accepted 9 March 2026. Date of publication 16 March 2026; date of current version 23 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62371359, in part by the Shaanxi Provincial Science and Technology Department under Grant 2024GX-ZDCYL-02-09, and in part by Xi'an Science and Technology Bureau under Grant 2024JHCLYB-0106. *(Corresponding author: Haitao Xu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3674175

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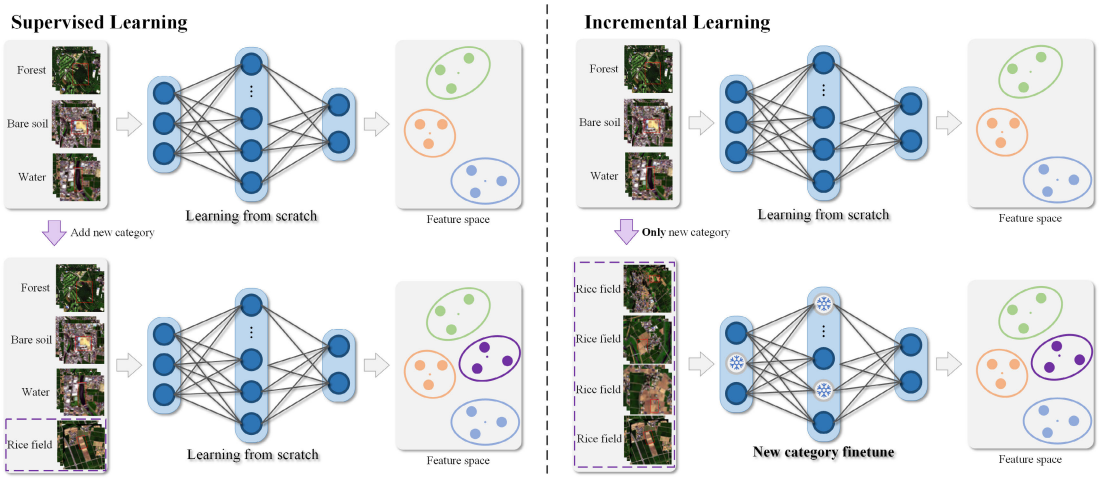


Fig. 1. Comparison of supervised learning and incremental learning. (Left) Supervised learning (from scratch): it requires retraining the entire model with both old and new categories when new data arrives, leading to high computational costs. (Right) Incremental learning: only new category data are used to update the model while preserving existing knowledge, achieving efficient adaptation without catastrophic forgetting.

mation and an information aggregation constraint (IAC) to improve performance by redistributing global information across the feature maps of different modalities.

Concurrently, several multilayer perceptron (MLP)-based networks have been proposed for single-modal or multimodal RSI classification [19], [20], which replace Transformers with MLPs due to their lighter and more compact architecture. Additionally, Mamba [21], [22], [23], a novel framework, is gaining prominence for its ability to harmonize global receptive fields with computational efficiency, which has been further extended to MRSI classification tasks [24], [25], [26]. Specifically, Liao et al. [24] proposed a multimodal Mamba fusion module (MMFM), which effectively models long-range dependencies within multimodal sequences by capturing intrinsic modality-specific features and intermodal relationships, thereby enhancing fusion performance. In addition, Li and Huang [25] proposed an adaptive feature alignment network (AFA–Mamba), which utilizes a global–local Mamba framework to model contextual information through a 2-D selective scanning mechanism while introducing local bias to strengthen the spatial representation of local objects. Moreover, Zhang et al. [26] developed S2CrossMamba, where the inverted bottleneck cross-Mamba (IBCM) structure is designed to enrich feature representations by effectively integrating multimodal features through the inverted bottleneck architecture.

Despite these advancements, most existing methods remain constrained by static category paradigms, failing to adapt dynamically to emerging land cover classes in real-world remote sensing scenarios. Traditional approaches, when incrementally updated to evolving class distributions without mechanisms to preserve historical data patterns, often suffer from catastrophic forgetting of previously acquired knowledge. To mitigate this, they typically require complete retraining from scratch, which is computationally expensive and impractical for dynamic real-world scenarios. To address this critical challenge, incremental learning [27], [28], [29] has emerged as a promising solution, enabling models to continuously acquire new knowledge while preserving previously learned information. As shown in Fig. 1, supervised learning requires costly retraining on all data (old and new categories), whereas incremental learning efficiently updates the model using only new category data while retaining prior knowledge. Notably, recent advances in prompt-based incremental learning have demonstrated notable success in adapting models to new classes with minimal retraining. For instance, prompt-tuning methods [30], [31] introduce learnable prompts to guide the model in retaining prior knowledge while adapting to new classes, thereby alleviating catastrophic forgetting. In particular, while prompt-based methods offer a promising direction, existing approaches employ generic prompt tuning strategies that are not expressly designed for the heterogeneous and complementary nature of multimodal remote sensing data.

To bridge this gap, we propose a novel dual-flow prompt-expert Mamba (DF-PEM) network for incremental classification (IC) of MRSI. Specifically, inspired by mixture of experts (MoE), the innovative prompt-driven assistant-expert (PAE) mechanism coordinates incremental knowledge assimilation and anti-forgetting preservation implemented through paired assistant-expert prompts. Employing prompt assistants as adaptive gating controllers, DF-PEM activates specific experts through feature-affinity routing across both historical and emerging categories so that all accumulated experts for both old and new categories remain persistently stored. The training paradigm of DF-PEM mainly consists of two stages: the initial training phase and the incremental learning phase. During the initial training phase, the DF-PEM is trained on samples from initial categories in a supervised manner, and a dedicated spherical Gaussian sift (SGS) strategy is exploited to generate initial prototypes, which models the complex feature distribution of MRSI with a spherical Gaussian distribution to mitigate the impact of noise and feature outliers. It captures the potential centers of the feature space, thereby generating more representative prototypes for better preserving knowledge of the old categories. Then, in the incremental learning phase, new category samples are utilized to dynamically update the

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prompts and the partially frozen backbone, enabling the model to dynamically assimilate novel knowledge while preliminarily retaining its accumulated vintage knowledge. Building upon this, to provide enhanced mitigation against catastrophic forgetting, the prototype-aware regularization classifier (PRC) is introduced to adaptively derive the regularization coefficients. These coefficients strategically constrain parameter updates to mitigate prediction bias drift and further enforce robust equilibrium in the retention of both vintage and new knowledge.

The primary contributions of our work are outlined as follows.

1) In this article, we propose a novel DF-PEM for MRSI classification, integrating a PAE mechanism with Mamba to maintain a memory space to mitigate catastrophic forgetting.
2) An SGS strategy is introduced during the initial training phase, which models the complex feature distributions of old categories and eliminates the effects of noise and feature outliers to generate effective representative prototypes.
3) To further ensure the robust retention of both vintage and new knowledge, the PRC is designed for the incremental learning phase, specifically to balance the prediction bias of the model toward the new categories through adaptive optimization of stage-wise regularization coefficients.

II. Related Work

A. Class-Incremental Learning

Class-incremental learning (CIL) [27], [32] is an effective strategy for addressing new tasks while alleviating catastrophic forgetting of previously acquired knowledge. IC, a form of adaptive classification with incremental learning, refers to classifiers that, once initially trained, are capable of recognizing new classes without requiring full retraining of the entire system during subsequent incremental learning phases. Within the current research works on CIL, there are three main categories: regularization-based [33], [34], examplar-based [28], [35], and bias correction [27]. First, as a regularization-based method, the learning without forgetting (LwF) model [34] trains the network exclusively on new data while preserving the original recognition capability. Second, exemplar-based methods selectively incorporate a small subset of old samples into the learning of new knowledge, utilizing a sample replay strategy. For instance, iCaRL [35] proposed a class-incremental learning strategy that enables the progressive addition of new classes while simultaneously learning strong classifiers and data representations. Finally, the BiC [27] proposes a scalable framework for incremental learning, enabling efficient handling of large datasets by progressively learning new classes without catastrophic forgetting by adjusting the classifier's bias toward previously learned classes.

Recently, incremental learning has been applied to unimodal remote sensing classification [36], [37], [38], [39]. Specifically, CILEA-NET [37] employs a curriculum learning strategy to tackle IC for remote sensing data. The order in which new classes are introduced is strategically designed based on their similarity to previously trained classes. Bai et al. [38] proposed a linear programming-based incremental learning classifier that enables the acquisition of new capabilities by leveraging a pretrained model and incorporating a single sample from a new class. Although the aforementioned methods have yielded promising results, the IC of multimodal remote sensing imagery remains an area in urgent need of further research and exploration. In this work, we explore the IC of multimodal remote sensing images and propose incorporating prompt-based techniques alongside classifier correction to enhance model performance.

B. Mamba-Based Works

Mamba [21], an advanced foundational model, has recently garnered significant attention due to its various implementations across a range of visual tasks, including Vision Mamba (ViM) [22], LocalMamba [40], and PlainMamba [41]. Specifically, ViM [22] is designed for vision tasks, where it utilizes position embeddings to mark image sequences and compresses the visual representation with bidirectional state space models before processing them through the Mamba encoder. Furthermore, LocalMamba [40] introduces a novel local scanning strategy that divides images into distinct windows, effectively capturing local dependencies while maintaining a global perspective. Subsequently, GrootVL [42] is proposed to enhance long-range modeling for both vision and language tasks by incorporating an input-aware, tree-based topological structure.

Due to the strengths of Mamba, it has been increasingly adapted to the remote sensing domain, with numerous related works being proposed [43], [44], [45], [46]. For example, RS-Mamba [43] is proposed for dense prediction tasks on large very-high-resolution remote sensing imagery, which incorporates an omnidirectional selective scan module to effectively process images from multiple spatial directions. Then, SpectralMamba [44] has been proposed for hyperspectral image classification, which simplifies spectral dynamics modeling through spatial–spectral encoding and input-dependent hidden state operations to address computational inefficiencies. Furthermore, Mamba-in-Mamba architecture is designed for hyperspectral image classification, combining a novel centralized Mamba-cross-scan mechanism, a tokenized Mamba encoder with Gaussian decay mask and semantic token fuser, and a weighted MCS Fusion module with multiscale loss design to enhance feature aggregation, training efficiency, and classification performance. Besides, MaDiNet [47] addresses the challenge of SAR target detection under complex noncooperative environments (e.g., sparse targets and background interference) by proposing a gamma diffusion model network integrated with a MambaSAR module, which models SAR image statistics and captures target spatial structural information. CD-Lamba [48] addresses the poor locality perception of existing Mamba-based remote sensing change detection methods, proposing a locally adaptive state space model with local enhancement and cross-temporal fusion. Despite the advancements in various Mamba architectures for remote sensing tasks, the application of Mamba to multimodal remote sensing image classification remains underexplored. In this study, we propose a novel approach that leverages Mamba's

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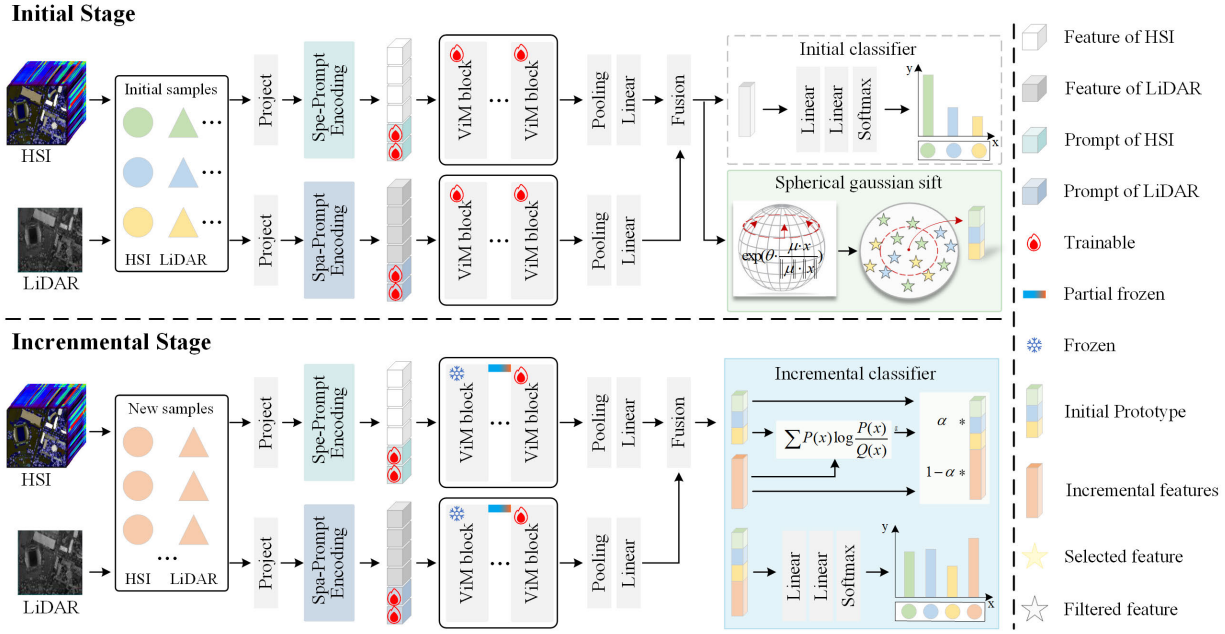


Fig. 2. Flowchart of the proposed DF-PEM, which consists of two distinct phases. During the initial phase, the PEM is trained in a supervised manner using initial samples, with feature prototypes being simultaneously acquired through SGS. In the incremental phase, a partially frozen network undergoes fine-tuning with new samples, while the classification outcome is ultimately produced by a PRC that integrates prior knowledge with incremental learning.

strengths for multimodal data, offering an effective solution for incremental learning in MRSI classification.

III. Methods

The overall architecture of the DF-PEM, as depicted in Fig. 2, operates through a dual-stage paradigm comprising an initial stage and an incremental stage while integrating three core components: the PAE mechanism, the SGS strategy, and the PRC. In the initial stage, the training samples from the predefined base classes jointly optimize the ViM backbone and the learnable prompts. Concurrently, the SGS strategy derives initial prototypes that encapsulate prior knowledge distributions, establishing robust prototype representations to support subsequent incremental learning. During the incremental stage, novel class samples fine-tune partially frozen ViM layers and the prompts, dynamically consolidating new patterns while preserving vintage knowledge to mitigate catastrophic forgetting. Furthermore, the PRC adaptively optimizes regularization coefficients between emerging features and initial prototypes, balancing prediction biases toward new categories. Subsequently, Section III-A elaborates on the dual-stage architecture, and Sections III-B–III-D illustrate the tripartite components through formal mathematical formulations and algorithmic details.

A. Dual-Stage Training

The training for DF-PEM comprises two distinct phases: the initial stage and the incremental stage, operating on a Multimodal training dataset $D_{train} = D_{ini} \cup D_{incre}$. Here, $D_{ini} = \{x_{ini}^{spe}, x_{ini}^{spa}, y_{ini}\}$ denotes the base-class data with labels $y_{ini} \in C_{base}$, while $D_{incre} = \{x_{incre}^{spe}, x_{incre}^{spa}, y_{incre}\}$ represents novel class samples where $y_{incre} \in C_{novel}$.

Specifically, during the initial stage, the network is preliminarily trained with samples from D_{ini}

$$y^* = f_{cls}\left(f_{fusion}\left(f_{spa}\left(x_{ini}^{spa}\right), f_{spe}\left(x_{ini}^{spe}\right)\right)\right) \tag{1}$$

where x_{ini}^{spa} and x_{ini}^{spe} represent spatial-dominant modality and spectral-dominant modality, $f_{spa}(\bullet)$ and $f_{spe}(\bullet)$ denote the dual-branch feature extractor, $f_{fusion}(\bullet,\bullet)$ represents the fusion module for the features of two modalities, and $f_{cls}(\bullet)$ is the initial classifier. For effective multimodal fusion, we adopt a lightweight linear layer to map the spectral features (from the HSI branch) and spatial features (from the LiDAR/SAR branch) into a unified feature space. This step eliminates the heterogeneity between different modal features, ensuring spectral and spatial information can be directly compared and fused. The integrated features generated thereby simultaneously carry spectral material information and spatial structure information, laying a foundation for accurate classification.

Furthermore, the initial prototype representations z_{proto} are constructed through the SGS during the initial training phase. The classification loss and overall loss of the initial stage are as follows:

$$\mathcal{L}_{cls} = -\sum_{i=1}^{N} y_i log\left(y_i^*\right) \tag{2}$$

$$\mathcal{L}_{init} = \mathcal{L}_{cls} + \mathcal{L}_{PE} \tag{3}$$

where $\mathcal{L}_{init}$ represents the overall loss, $\mathcal{L}_{cls}$ denotes the classification loss, and $\mathcal{L}_{PE}$ stands for the prompt encoding loss that will be discussed in detail later.

In the incremental stage, the novel samples from D_{incre} are utilized to train the partially frozen network

$$z^* = f_{fusion}\left(f_{spa}\left(x_{incre}^{spa}\right), f_{spe}\left(x_{incre}^{spe}\right)\right) \tag{4}$$

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$$y^* = f'_{cls}\left(z^*, z_{proto}\right) \tag{5}$$

where f'_{cls} represents the incremental classifier that balances the retention of both vintage and new knowledge. Notably, the optimization objective in the incremental stage includes one additional term $\mathcal{L}_{align}$ compared to that in the initial stage

$$\mathcal{L}_{align} = 1 - \frac{z^* \cdot z_{proto}}{\|z^*\| \ \left\|z_{proto}\right\|} \tag{6}$$

$$\mathcal{L}_{incre} = \mathcal{L}_{cls} + \mathcal{L}_{PE} + \mathcal{L}_{align} \tag{7}$$

where $\mathcal{L}_{align}$ represents the feature alignment loss, which maintains a similar distribution as the initial prototypes in the feature space.

The following sections provide a detailed introduction to the tripartite components.

B. Prompt-Expert Mamba

To address the catastrophic forgetting in incremental learning, we propose a PAE mechanism integrated with the Mamba architecture, which dynamically activates specific experts for different categories to extract discriminative features. Compared with traditional MoE methods with generic gating, our PAE mechanism enables modality-category-specific expert activation for incremental learning, achieving a better balance between new category adaptation and old knowledge retention. We select the optimal expert via cosine similarity between input features and modality-specific prompt assistants and dynamically add new combinations of expert pairs for new categories without updating old ones, ensuring efficient and targeted expert utilization. The feature extractor of DF-PEM adopts a dual-branch architecture: one branch is dedicated to hyperspectral data and the other to LiDAR data. Both branches share the same network structure while addressing different dimensions, spectral for HSI and spatial for LiDAR. The architecture of the PEM is shown in Fig. 3. First, the training samples are flattened and projected through a projection layer to get the feature embedding $Q \in \mathbb{R}^{N\times d}$. Then, a PAE mechanism is designed, which leverages the gating role of prompt assistants to independently select the top-K experts for each feature embedding. Notably, the prompt assistants and prompt experts exhibit a one-to-one pairing relationship $\{(A_1, E_1), (A_2, E_2), (A_3, E_3), \ldots\}, A, E \in \mathbb{R}^d$, ensuring precise collaborative interactions within the mechanism. Specifically, the feature embeddings directly interface with the prompt assistants, which route the input samples to the relevant expert prompts

$$\begin{aligned} \mathcal{I}_{\text{topk}} &= \underset{i\in\{1,\ldots,N\}}{\text{Topk}}\ (SIM(F, A_i)) \\ \mathcal{E}_{\text{selected}} &= \left\{E_j \mid j \in \mathcal{I}_{\text{topk}}\right\} \end{aligned} \tag{8}$$

where $\mathcal{I}_{\text{topk}}$ denotes the selected top-k index for query Q, SIM denotes similarity calculation (cosine similarity), and $\mathcal{E}_{\text{selected}}$ stands for the selected prompt experts. Then, the feature embedding and the selected prompt experts are concatenated

$$\mathbf{S}_0 = \text{concat}\,(\mathbf{Q}, \mathcal{E}) \tag{9}$$

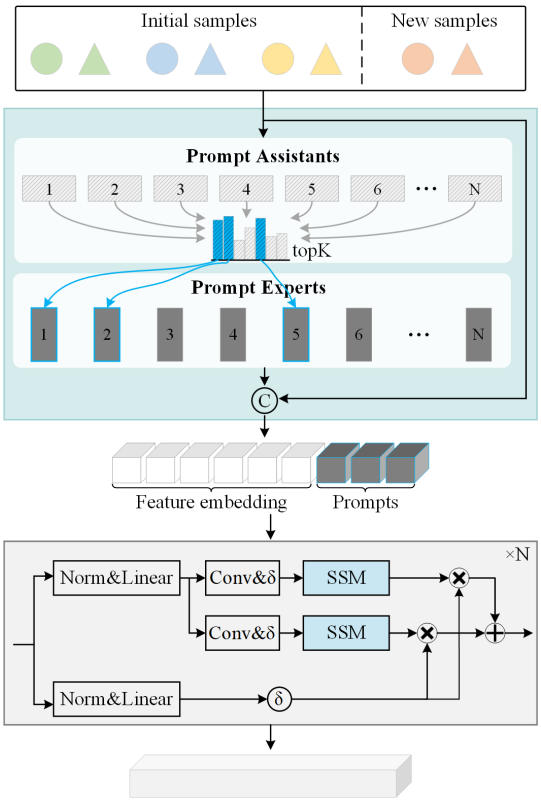


Fig. 3. Architecture of the PEM. Feature embeddings from initial or new samples are first processed by prompt assistants, where a gating mechanism selects the top-k experts. These selected expert features are subsequently concatenated with the original embeddings and routed to the ViM blocks for further processing.

where $\mathbf{S}_0 \in \mathbb{R}^{(N+K)\times d}$ is subsequently propagated through the ViM blocks [30] for feature extraction, thereby enabling synergistic integration of current context with historical knowledge priors. The specific calculation process of the ViM block is as follows:

$$y^*_{forward} = f_{forward}\,(S_{l-1}) \tag{10}$$

$$y^*_{backward} = f_{backward}\,(S_{l-1}) \tag{11}$$

$$S_l = Linear\left(y^*_{forward} + y^*_{backward}\right) + S_{l-1} \tag{12}$$

where $f_{forward}$ and $f_{backward}$ denote the operations of the SSM branch, differing only in the processing direction. Notably, the prompts in the dual branches are modality-specific without parameter sharing. The HSI branch is configured with spectral prompts, which are optimized to recognize spectral patterns. The LiDAR/SAR branch uses spatial prompts, tailored to identify spatial structure features. This design ensures each branch's features are "dimension-pure," avoiding cross-modal interference and maximizing the uniqueness of each modality's contribution.

Besides, an additional term is incorporated into the loss function to pull the selected assistants $\mathbf{A}_{select}$ closer to their corresponding query features $\mathbf{Q}$, and the specific formula is as follows:

$$\mathcal{L}^{hsi}_{PE} = 1 - \frac{\mathbf{A}^{hsi}_{select} \cdot \mathbf{Q}^{hsi}}{\left\|\mathbf{A}^{hsi}_{select}\right\| \ \left\|\mathbf{Q}^{hsi}\right\|} \tag{13}$$

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$$\mathcal{L}_{PE}^{lidar} = 1 - \frac{\mathbf{A}_{select}^{lidar} \cdot \mathbf{Q}^{lidar}}{\left\|\mathbf{A}_{select}^{lidar}\right\| \ \left\|\mathbf{Q}^{lidar}\right\|} \tag{14}$$

$$\mathcal{L}_{PE} = \mathcal{L}_{PE}^{hsi} + \mathcal{L}_{PE}^{lidar}. \tag{15}$$

As the loss decreases, the selected prompt assistants become increasingly closer to their corresponding query features.

C. Spherical Gaussian Sift

In the initial stage, to memorize the vintage knowledge, we propose the SGS strategy, a prototype-oriented feature selection method designed to preserve critical knowledge during model initialization. By projecting features onto a hyperspherical manifold and employing Gaussian density estimation, SGS identifies high-confidence regions in the latent space, filtering out noisy or outlier samples that may destabilize historical knowledge retention.

The initial prototype center $\boldsymbol{\mu}^0 \in \mathbb{R}^d$ is computed as follows, which represents the empirical mean of all features in the latent space:

$$\begin{aligned} \mathbf{z}_i &= f_{fusion}\left(f_{spa}\left(x_{ini}^{spa}\right), f_{spe}\left(x_{ini}^{spe}\right)\right) \\ \boldsymbol{\mu}^0 &= \frac{1}{N}\sum_{i=1}^{N}\mathbf{z}_i \end{aligned} \tag{16}$$

where N represents all the training samples of the initial stage. Then, for a feature $\mathbf{z}_i \in \mathbb{R}^d$, the probability of it belonging to the spherical Gaussian distribution centered at u^0 is defined as

$$P\left(\mathbf{z}_i; \boldsymbol{\mu}^0, \theta\right) = exp\left(\theta \cdot \left(\frac{\mathbf{z}_i \cdot \boldsymbol{\mu}_0}{\|\mathbf{z}_i\| \cdot \left\|\boldsymbol{\mu}_0\right\|}\right)\right) \tag{17}$$

where θ is the temperature parameter controlling the concentration of the distribution. Notably, this probability function assigns higher values to features closer to the prototype center on the hypersphere. The dissimilarity score initially is

$$dis_i = 1 - P\left(\mathbf{z}_i; \boldsymbol{\mu}^0, \theta\right). \tag{18}$$

Then, rank features by ascending dissimilarity and select top-K features with minimal dis_i

$$\begin{aligned} SortedIndices &= argsort_{i\in 1,\ldots,N}\left(dis_i\right) \\ S_{selected} &= \{\mathbf{z}_i | i \in SortedIndices\,[:K]\} \end{aligned} \tag{19}$$

where $K = \lfloor \rho N \rfloor$ and $\rho \in [0, 1]$ represents the selection ratio. Therefore, the refined prototype center $\boldsymbol{\mu}_{refine}$ is computed as the mean of the selected features

$$\boldsymbol{\mu}_{refine} = \frac{1}{K}\sum_{S_{selected}}\mathbf{z}_i. \tag{20}$$

D. Incremental Classifier

In the incremental stage, to further mitigate catastrophic forgetting, the PRC is introduced to adaptively learn the regularization coefficients, balancing model prediction bias. Specifically, to address the inherent bias toward novel classes during the incremental stage, Jensen–Shannon divergence (JSD)-driven regularization mechanism that dynamically calibrates the model's adaptation to new samples while preserving knowledge from historical prototypes.

TABLE I
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE HOUSTON DATASET (HSI+LIDAR)

Class	Name	Number of Samples
1	Health grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	196
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking lot1	1233
13	Parking lot2	469
14	Tennis court	430
15	Running track	660
Total		15029

For each novel class sample $\mathbf{x}_i^{new}$, the multimodal feature extractor and fusion module generate a feature embedding $\mathbf{z}_i^{new} \in \mathbb{R}^d$. Then, define its similarity distribution over initial prototype.

To quantify the distribution shift between the novel sample and historical knowledge, compute the JSD between $\mathbf{z}_i^{new}$ and the initial prototype $\boldsymbol{\mu}_{refine}$

$$\alpha_i = \delta\left(\frac{1}{2}D_{\mathrm{KL}}\left(\mathbf{q}_i^{\mathrm{new}} \ \|\right) + \frac{1}{2}D_{\mathrm{KL}}\left(u\left\|\frac{\mathbf{q}_i^{\mathrm{new}} + u}{2}\right.\right)\right) \tag{21}$$

where D_{KL} denotes the Kullback–Leibler divergence and δ represents the sigmoid function. Then, the final embeddings are computed

$$\mathbf{z}_i^{final} = concat\left(\alpha_i * \mathbf{u}_{refine}, (1-\alpha_i) * \mathbf{z}_i^{new}\right). \tag{22}$$

The final predicted probability is computed by passing the representation through two linear layers and a softmax function

$$y^* = Softmax\left(Linear\left(Linear\left(\mathbf{z}_i^{final}\right)\right)\right). \tag{23}$$

IV. EXPERIMENTS

A. Datasets Description

To rigorously evaluate the robustness and generalizability of the proposed DF-PEM in complex multimodal scenarios, we conduct experiments on three benchmark datasets: the University of Houston dataset (2013) (HSI+LiDAR), MUUFL Gulfport dataset (2013) (HSI+LiDAR), and Trento dataset (2021) (HSI+LiDAR). These datasets represent diverse geographic environments and sensor configurations, ensuring comprehensive validation under heterogeneous conditions. The detailed enumeration of the classes and the corresponding sample counts for each dataset, as summarized in Tables I–III.

1) *Houston:* As the standard benchmarking dataset from the 2013 IEEE GRSS Data Fusion Contest, the Houston dataset contains co-registered HSI and LiDAR-derived digital surface model (DSM) acquired over the University of Houston campus and its adjacent urban areas.

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TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE MUUFL DATASET (HSI+LIDAR)

Class	Name	Number of Samples
1	Trees	23246
2	Mostly grass	4270
3	Ground surface	6882
4	Dirt	1862
5	Road	6687
6	Water	466
7	Building shadow	2233
8	Building	6240
9	Sidewalk	1385
10	Yellow curb	183
11	Cloth panels	269
Total		53732

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE TRENTO DATASET (HSI+LIDAR)

Class	Name	Number of Samples
1	Trees	4034
2	Buildings	2903
3	Ground	479
4	Woods	9123
5	Vineyard	10501
6	Roads	3647
Total		30214

This dataset provides fine-grained spectral-spatial characterization for 15 urban land cover categories, with a total of 15 029 annotated samples. The detailed class distribution is documented in Table I.

2) *MUUFL:* Developed by the University of Southern Mississippi, this multimodal dataset focuses on complex urban scenes in Gulfport, Mississippi. It synergistically combines high-resolution hyperspectral cubes with LiDAR point cloud data, specifically designed to facilitate urban classification and object detection tasks. The dataset encompasses 11 distinct object categories with 53 732 validated ground truth samples, whose statistical breakdown is presented in Table II.
3) *Trento:* Captured in the agricultural valleys of South Trento, Italy, this benchmark dataset integrates airborne hyperspectral scans with LiDAR topography data, targeting vegetation and crop type discrimination. Featuring 6 agriculturally significant land use classes with 30 214 georeferenced samples, its categorical composition is systematically organized in Table III.

B. Evaluating Indicator

The classification performance is quantitatively assessed using four widely adopted metrics: class-specific accuracy (CA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). These metrics are formally defined as follows:

$$\mathrm{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{\mathrm{TP}i}{N_i^{\mathrm{true}}} \tag{24}$$

$$\mathrm{OA} = \frac{\sum i = 1^{C}\mathrm{TP}i}{N^{\mathrm{total}}} \tag{25}$$

$$p_e = \frac{\sum i = 1^{C} N_i^{\mathrm{true}} \cdot N_i^{\mathrm{pred}}}{\left(N^{\mathrm{total}}\right)^2} \tag{26}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - p_e}{1 - p_e} \tag{27}$$

where C denotes the total number of semantic classes, TP_i represents the true positive samples for the ith class, N_i^{true} indicates the ground truth sample count of class i, and N_i^{pred} corresponds to the predicted sample count for class i. The term $N^{\mathrm{total}} = \sum_{i=1}^{C} N_i^{\mathrm{true}}$ defines the total number of samples in the dataset, ensuring the statistical consistency between class-specific measurements and global accuracy evaluation.

C. Implementation Details

All experiments are performed on a computer equipped with Pytorch, Ubuntu system, 3.60-GHz CPU, GeForce RTX 2080 Ti GPU, and 64-GB memory. For PTFNet, adaptive moment estimation (Adam) is utilized as the optimization scheme, and the patch size of the input is 9 × 9 for HSI and 15 × 15 for LiDAR. All experiments were conducted on a computational platform running Ubuntu 20.04 LTS with PyTorch 1.9.0, configured with an Intel Core i9-10900K CPU @ 3.60 GHz, NVIDIA GeForce RTX 2080 Ti GPU (11-GB VRAM), and 64-GB DDR4 RAM.

For our DF-PEM implementation, we employed the Adam optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) with an initial learning rate of 0.001, utilizing spatial input patches of size 9×9 pixels for hyperspectral data and 15×15 pixels for LiDAR data. To ensure fair comparison, all baseline methods were rigorously adapted to our incremental learning framework while maintaining their original architectural parameters as specified in their respective publications. Besides, a stratified random sampling strategy was implemented, where 30 labeled samples per class were selected for training, with the remaining samples allocated to the test set. This allocation scheme maintains class distribution integrity while ensuring sufficient representation for minority classes during model optimization.

D. Comparing With Other Algorithms

Several MRSI classification algorithms are compared to verify the performance of the proposed DF-PEM, including SpectralFormer [49], SSMLP [19], AsyFFNet [50], DHViT [13], MLPMixer [20], Frit [14], SWFormer [51], and S2CrossMamba [26]. Notably, each network is integrated into our incremental learning framework to enable systematic comparison.

As documented in Table IV, the proposed DF-PEM achieves state-of-the-art performance on the Houston 2013 benchmark, attaining 84.10% OA (the highest among all methods), 84.03% AA (the highest among all methods), and 0.8281 Kappa (the highest among all methods). Compared to the second-best methods, DF-PEM outperforms S2CrossMamba by 0.45% in OA (83.65%), SSMLP by 0.12% in AA (83.91%), and S2CrossMamba by 0.0044 in Kappa (0.8237). This tri-metric

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TABLE IV
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE HOUSTON DATASET

Houston	SpectralFormer	SSMLP	AsyFFNet	DHViT	MLPMixer	FriT	SWFormer	S2CrossMamba	DF-PEM
1	**98.44**	95.74	81.82	88.94	95.33	92.22	93.86	96.40	93.86
2	96.49	**96.73**	72.47	78.02	86.93	87.66	91.99	94.28	92.48
3	95.65	99.10	93.55	**99.85**	97.75	88.31	95.20	95.17	95.95
4	90.61	94.98	86.66	71.91	95.39	90.36	98.16	97.53	**98.19**
5	93.73	**99.67**	90.84	93.23	91.17	93.23	69.55	96.29	92.74
6	84.75	87.46	83.05	83.73	86.10	87.12	84.41	**87.80**	84.41
7	**91.92**	87.72	72.05	77.22	88.37	81.91	83.20	85.62	84.49
8	67.87	73.15	61.20	70.59	75.54	75.54	83.61	56.59	**83.94**
9	**85.35**	75.53	43.13	69.15	69.80	64.98	71.28	80.44	71.52
10	81.04	69.84	44.53	82.54	74.69	**82.87**	63.41	63.17	62.82
11	45.31	66.80	63.90	85.48	71.87	72.45	79.42	75.44	**80.50**
12	68.58	73.23	81.55	75.98	80.05	77.97	78.22	**87.86**	79.55
13	66.51	93.39	87.02	74.72	54.90	**95.67**	68.56	86.81	73.35
14	97.99	85.68	**98.74**	93.22	97.74	10.80	87.69	95.22	91.46
15(new)	**93.65**	59.68	49.68	64.13	73.97	86.03	90.00	87.46	75.24
OA(%)	83.17	83.54	71.62	79.91	82.86	80.97	82.18	83.65	**84.10 (↑0.45)**
AA(%)	83.86	83.91	74.01	80.58	82.64	79.14	82.57	83.79	**84.03 (↑0.24)**
Kappa	81.79	82.19	69.58	78.36	81.50	79.43	80.75	82.37	**82.81 (↑0.44)**

TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE MUUFL DATASET

MUUFL	SpectralFormer	SSMLP	AsyFFNet	DHViT	MLPMixer	FriT	SWFormer	S2CrossMamba	DF-PEM
1	79.07	74.78	86.56	79.99	**88.53**	83.88	86.47	87.42	88.11
2	**68.68**	64.01	58.04	36.84	30.12	50.97	25.26	34.32	42.64
3	58.36	**62.62**	11.47	28.08	32.94	42.29	33.99	49.45	50.67
4	83.85	87.36	35.80	86.41	73.22	67.98	**90.20**	81.12	69.77
5	66.89	66.97	21.29	64.11	86.83	79.10	**92.26**	85.28	87.89
6	31.19	41.74	25.23	96.79	98.17	78.44	97.94	**98.85**	91.06
7	91.92	**94.55**	76.08	82.52	77.49	85.75	83.66	85.61	82.80
8	78.10	84.64	94.11	95.60	94.07	66.88	89.24	95.41	**96.38**
9	63.17	**67.82**	36.01	17.71	46.42	59.78	56.53	47.05	58.89
10	83.66	**91.50**	26.80	72.55	89.54	81.70	80.39	73.01	63.40
11(new)	64.02	85.36	15.06	82.85	88.70	76.99	64.44	**89.16**	80.75
OA(%)	73.81	73.42	62.97	68.60	75.22	72.17	75.14	76.76	**78.97 (↑2.21)**
AA(%)	69.90	73.67	44.22	67.59	73.27	70.35	72.76	73.15	**73.85 (↑0.18)**
Kappa	66.98	67.04	54.04	60.53	68.55	64.75	68.55	70.47	**73.00 (↑2.53)**

TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON THE TRENTO DATASET

Trento	SpectralFormer	SSMLP	AsyFFNet	DHViT	MLPMixer	FriT	SWFormer	S2CrossMamba	DF-PEM
1	85.86	98.38	78.35	98.03	96.80	96.63	97.90	98.27	**98.43**
2	67.66	23.53	96.80	95.86	97.13	89.14	92.79	98.08	**98.47**
3	71.05	87.51	46.10	48.04	91.00	59.91	60.36	75.95	**97.33**
4	92.00	98.63	90.08	**100.00**	99.99	**100.00**	99.91	**100.00**	**100.00**
5	98.28	99.80	94.85	97.69	98.00	98.39	**99.86**	99.14	99.21
6(new)	84.22	64.31	61.55	62.60	83.91	**85.49**	83.40	66.95	81.71
OA(%)	89.92	88.06	87.18	93.39	97.08	95.83	96.77	96.02	**97.41 (↑0.33)**
AA(%)	83.18	78.73	77.95	78.70	94.92	88.26	89.21	90.30	**95.86 (↑0.94)**
Kappa	86.51	84.01	83.13	91.13	96.10	94.42	95.68	94.68	**96.55 (↑0.45)**

superiority demonstrates our method's balanced capability in handling Houston's complex urban-landcover heterogeneity.

As shown in Table V, the proposed DF-PEM establishes new state-of-the-art performance on the MUUFL urban scene dataset, achieving the highest OA (78.97%) and Kappa (0.7300) among all compared methods. For the AA metric, our method obtains 73.85%, slightly outperforming SSMLP (73.67%) by 0.18%. These results demonstrate that DF-PEM delivers superior overall classification accuracy while maintaining robust performance across individual categories.

As quantitatively validated in Table VI, the proposed DF-PEM demonstrates superior classification performance on the Trento dataset, achieving state-of-the-art results across all three holistic metrics: 97.41% OA (highest among all methods), 95.86% AA (highest among all methods), and 96.55% Kappa (highest among all methods). Compared to the second-best

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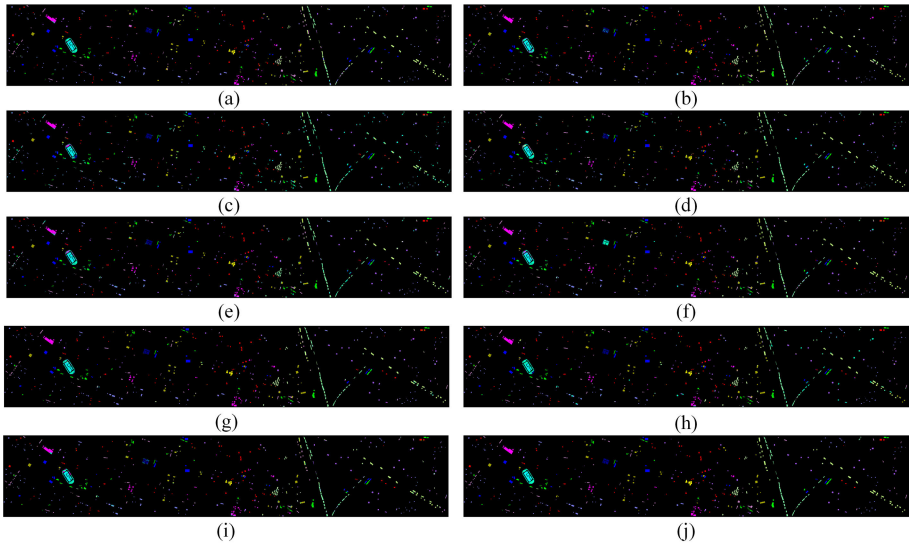


Fig. 4. Classification maps of the Houston dataset. (a) SpectralFormer. (b) SSMLP. (c) AsyFFNet. (d) DHViT. (e) MLPMixer. (f) FriT. (g) SWFormer. (h) S2CrossMamba. (i) DF-PEM. (j) Groundtruth.

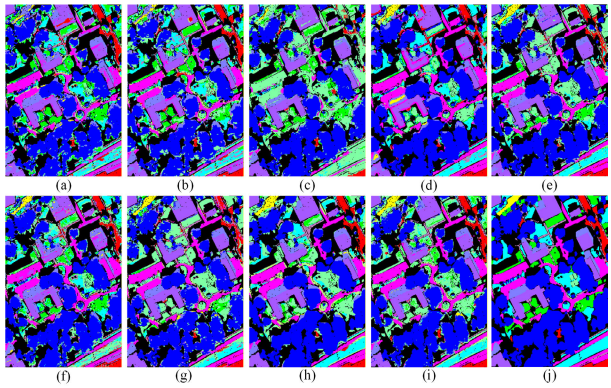


Fig. 5. Classification maps of the MUUFL dataset. (a) SpectralFormer. (b) SSMLP. (c) AsyFFNet. (d) DHViT. (e) MLPMixer. (f) FriT. (g) SWFormer. (h) S2CrossMamba. (i) DF-PEM. (j) Groundtruth.

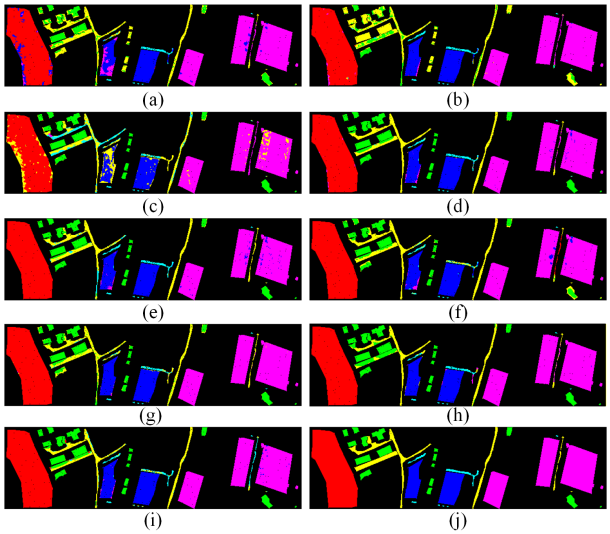


Fig. 6. Classification maps of the Trento dataset. (a) SpectralFormer. (b) SSMLP. (c) AsyFFNet. (d) DHViT. (e) MLPMixer. (f) FriT. (g) SWFormer. (h) S2CrossMamba. (i) DF-PEM. (j) Groundtruth.

methods, DF-PEM improves OA by 0.33% over MLPMixer (97.08%), AA by 0.94% over MLPMixer (94.92%), and Kappa by 0.45% over MLPMixer (96.10%). At the class level, DF-PEM attains the highest accuracy in 3 out of 6 categories (Classes 1–4) while achieving competitive performance in the remaining categories.

Combined with the experimental results (Tables IV–VI) and dataset characteristics (Tables I–III), high misclassification mainly occurs in original small-sample classes (e.g., Houston 7, MUUFL 10, and Trento 3), new incremental classes, and spectrally and spatially similar classes. Even with 30 labeled samples per class, original small-sample classes still suffer from inadequate feature learning due to insufficient training sample representativeness, exacerbated by similar-class interference and data noise. DF-PEM alleviates such issues better than baseline models via the PAE and SGS mechanisms, demonstrating stronger robustness.

As illustrated in Figs. 4–6, the proposed DF-PEM demonstrates superior performance in classification results that exhibit the closest resemblance to ground truth. Particularly noteworthy is its exceptional performance on the challenging MUUFL dataset with complex terrain characteristics, where our method achieves remarkable boundary delineation accuracy while maintaining reduced noise interference. These observations substantiate that the synergistic combination of our novel PAE mechanism and the PRC architecture effectively facilitates both the acquisition of emerging class representations and the preservation of historical knowledge. The framework's dual capability enables robust feature adaptation for new categories while preventing catastrophic forgetting of previously learned information, as evidenced by the improved edge definition and spatial consistency in the generated classification maps compared to comparative methods.

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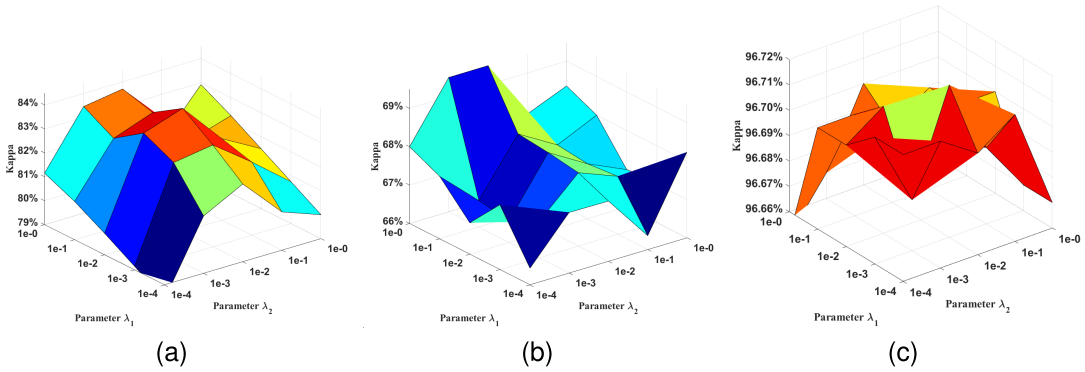


Fig. 7. Parameter tuning of λ_1 and λ_2 for the proposed DF-PEM on the three experimental datasets. The x-axis and y-axis represent λ_1 and λ_2, respectively, while the z-axis denotes the OA. (a) Houston. (b) MUUFL. (c) Trento.

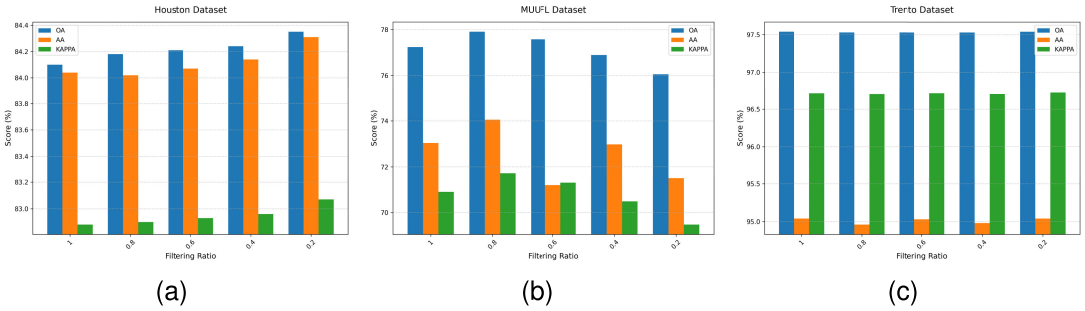


Fig. 8. Parameter tuning of the ratio of SGS on three datasets. For each subplot—(a) Houston, (b) MUUFL, and (c) Trento—the x-axis shows the ratio values 0.2, 0.4, 0.6, 0.8, and 1.0, and the y-axis shows accuracy. At each ratio, three bars represent OA (blue), AA (orange), and Kappa coefficient (green).

E. Hyperparameter Analysis of Loss Function Components

In our experiments, the total loss function consists of three parts: $\mathcal{L}_{incre} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{PE} + \lambda_1 \mathcal{L}_{align}$, where λ_1 and λ_2 are the weights of $\mathcal{L}_{PE}$ and $\mathcal{L}_{align}$. Given the divergent characteristics of benchmark datasets in terms of data distribution shift and incremental learning difficulty, we conducted comprehensive per-dataset hyperparameter optimization through an exhaustive five-order magnitude grid search [predefined value ranges for λ_1 and λ_2 ($1e^{+0}$, $1e^{-1}$, $1e^{-2}$, $1e^{-3}$, $1e^{-4}$)]. To better understand how the loss parameters work together, we created a 3-D color map (Fig. 7) showing how different combinations of λ_1 (horizontal axis) and λ_2 (vertical axis) affect Kappa (height axis). The results show clear patterns. For the Houston dataset, setting both parameters to 0.01 ($\lambda_1 = 10^{-2}, \lambda_2 = 10^{-2}$) gives the best accuracy (84.30%). The MUUFL dataset works best when λ_1 is set to 1 with $\lambda_2 = 0.01$ ($\lambda_1 = 10^0$ and $\lambda_2 = 10^{-2}$), achieving 73.00% accuracy. For Trento dataset, the optimal combination is $\lambda_1 = 0.001$ with $\lambda_2 = 0.01$ ($\lambda_1 = 10^{-3}$ and $\lambda_2 = 10^{-2}$) reaching 96.72% accuracy.

F. Hyperparameter Analysis of the Ratio of SGS

The parameter tuning process for the proposed PTFNet was systematically conducted by varying the ratio of spherical Gaussian sifting across three benchmark datasets: Trento, Houston, and MUUFL. As shown in Fig. 8, the relationship between parameter configurations and model performance was analyzed using three key metrics: OA, AA, and Kappa. For each dataset, the ratio values were tested at intervals of 1.0, 0.8, 0.6, 0.4, and 0.2.

Notably, the Trento dataset exhibited stable performance with OA consistently above 97.53% across all ratios, peaking at 97.54% for ratio = 0.2. The Houston dataset showed gradual improvement in OA from 84.10% to 84.35% as the ratio decreased, while MUUFL achieved optimal OA (77.91%) at 0.2. These results informed the final parameter selection, prioritizing configurations that balanced high OA with robust Kappa performance, ultimately adopting 0.2 for Trento and 0.8 for MUUFL.

Besides hyperparameter tuning, we further validated the rationality of the SGS strategy by conducting comparative experiments against two mainstream statistical modeling methods—kernel density estimation (KDE) and Gaussian mixture model (GMM)—across the three benchmark datasets. Across all datasets, the SGS strategy consistently outperforms both KDE and GMM in core classification metrics, including OA, AA, and the Kappa coefficient, demonstrating more reliable modeling of old-category feature distributions and better suppression of noise and outlier interference. Beyond classification performance, SGS also exhibits clear advantages in computational efficiency and stability: KDE suffers from high computational complexity that becomes prohibitive for large-scale remote sensing feature sets, while GMM relies on iterative optimization, which often leads to unstable prototype modeling for small-sample categories. In contrast, SGS achieves a more favorable tradeoff between modeling

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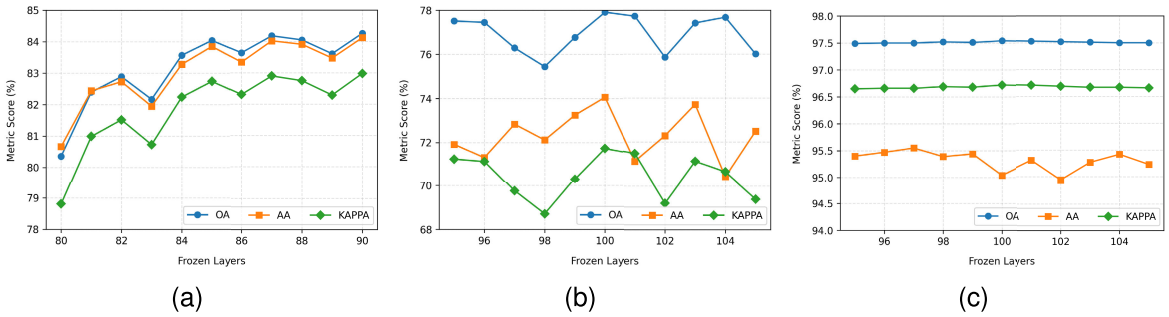


Fig. 9. Parameter tuning of frozen layers for feature extraction on three experimental datasets. For each subplot—(a) Houston, (b) MUUFL, and (c) Trento—the x-axis indicates the number of frozen layers, and the y-axis shows accuracy values. In each plot, three line plots represent OA (blue), AA (orange), and Kappa coefficient (green).

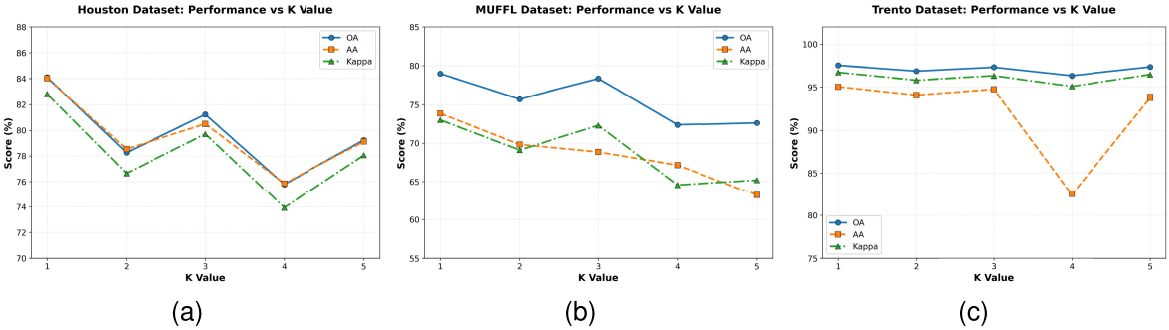


Fig. 10. Parameter tuning of the top-k experts on three datasets. For each subplot—(a) Houston, (b) MUUFL, and (c) Trento—the x-axis shows the K values 1, 2, 3, 4, 5, and the y-axis shows accuracy. At each ratio, three bars represent OA, AA, and Kappa coefficient.

accuracy, computational efficiency, and noise robustness, making it better suited for generating reliable old-category prototypes in multimodal remote sensing IC.

G. Hyperparameter Analysis of Frozen Layers

In the second phase, DF-PEM requires partial network freezing, and the optimal number of frozen layers may vary across different datasets. We therefore conducted hyperparameter experiments. Notably, we initially performed rough attempts to identify a suitable range for the number of frozen layers. Subsequently, within this determined range, we tested ten distinct freeze layer counts and plotted the experimental results in a line graph, as shown in Fig. 9. The optimal freeze layer counts are found to be 90 for the Houston dataset and 100 for the MUUFL dataset. The Trento dataset exhibits relatively stable performance, achieving comparable accuracy across the entire tested range.

H. Hyperparameter Analysis of K Value

As shown in Fig. 10, the Trento dataset exhibited robust performance with OA consistently above 96.3% across all K values, peaking at 97.54% for K = 1. The Houston dataset showed optimal OA (84.10%) and Kappa (82.81%) at K = 1, with a gradual decline to K = 4 followed by slight recovery at K = 5. MUUFL achieved its highest OA (78.97%) at K = 1, though both OA and Kappa showed more pronounced sensitivity to K-value variations compared to the other datasets. These results informed the final parameter selection, prioritizing configurations that balanced high OA with robust Kappa performance across all metrics. Considering the overall performance patterns and stability considerations, K = 1 was ultimately adopted for all three datasets, as it consistently delivered the strongest results across OA, AA, and Kappa metrics while minimizing performance variance.

TABLE VII
ABLATION COMPARISON OF EACH VARIANT OF DF-PEM

Dataset	Base	Prompts	SGS	OA(%)	AA(%)	Kappa
Houston	✓			82.23	81.78	80.20
	✓	✓		84.10	84.04	82.88
	✓		✓	82.61	82.44	81.22
	✓	✓	✓	**84.26**	**84.12**	**82.98**
MUUFL	✓			73.97	70.34	66.67
	✓	✓		77.24	73.04	70.89
	✓		✓	76.00	72.25	69.18
	✓	✓	✓	**77.91**	**74.05**	**71.72**
Trento	✓			96.43	94.64	95.24
	✓	✓		97.34	94.80	96.51
	✓		✓	97.20	94.84	96.27
	✓	✓	✓	**97.41**	**95.86**	**96.55**

I. Ablation Experiments

To validate the effectiveness of key components in DF-PEM, we conducted a comprehensive ablation study comparing different model variants: 1) Base (incremental framework + Mamba); 2) Base + Prompts; 3) Base + SGS module; and 4) Full DF-PEM (Base + Prompts + SGS). As summarized in Table VII, all three datasets consistently demonstrate performance improvements through component integration. For

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TABLE VIII
COMPREHENSIVE COMPARISON OF COMPUTATIONAL COMPLEXITY ACROSS DIFFERENT MODELS

Methods	Computational Cost	
	FLOPs (M)	Params (M)
SpectralFormer	5.93	0.23
SSMLP	1,011.00	1.23
AsyFFNet	18.66	0.57
DHViT	88.76	5.02
MLPMixer	7.25	1.52
FriT	2,077.00	24.34
SWFormer	24.92	4.40
S2CrossMamba	2207.00	**0.12**
DF-PEM (Ours)	**0.23**	0.31

the Houston dataset, the full implementation achieves optimal metrics (OA: 84.26%, AA: 84.12%, and Kappa: 82.98%), showing respective gains of 2.03%, 2.34%, and 2.78% over the base model. Similar trends are observed in MUUFL, where the complete framework improves OA by 3.94% and Kappa by 5.05% compared to the baseline. Notably, Trento exhibits remarkable stability and reaches peak performance (OA: 97.41%, AA: 95.86%, and Kappa: 96.55%) when incorporating all components.

J. Analysis of Parameter and FLOPs

Table VIII comprehensively compares the computational efficiency of different methods through FLOPs (floating-point operations) and parameter counts (Params). Notably, our proposed DF-PEM achieves an ultralightweight architecture with only 0.23M FLOPs (the lowest among all methods) and 0.31 M parameters, demonstrating superior overall computational efficiency compared to existing approaches.

Specifically, in terms of FLOPs, the following conditions hold.

1) DF-PEM's computational cost is 1–4 orders of magnitude lower than Transformer-based models, such as SpectralFormer (5.93 M, DF-PEM is 1/25.8 of its FLOPs), FriT (2077.00 M, 1/9030 of its FLOPs), and SWFormer (24.92 M, 1/108.3 of its FLOPs).
2) It is far more efficient than MLP-based models like SSMLP (1011.00 M, 1/4396 of its FLOPs) and MLPMixer (7.25M, 1/31.5 of its FLOPs).
3) Compared with the key baseline S2CrossMamba, DF-PEM's FLOPs (0.23 M) are merely 1/9596 of S2CrossMamba's 2207.00 M, representing a drastic reduction in computational overhead.

In terms of parameters, the following conditions hold.

1) While S2CrossMamba achieves the lowest parameter count (0.12M), its FLOPs (2207.00 M) are nearly 9596 times that of DF-PEM, leading to much higher computational consumption in practical inference.
2) DF-PEM's parameter count (0.31 M) is still more compact than most baselines, including SSMLP (1.23 M), AsyFFNet (0.57 M), DHViT (5.02 M), MLPMixer (1.52 M), FriT (24.34 M), SWFormer (4.40 M), and S2CrossMamba's parameter advantage is offset by its excessive FLOPs.

V. CONCLUSION

In this study, we address the critical challenge of catastrophic forgetting in multimodal remote sensing IC by proposing the DF-PEM framework. Unlike conventional fixed-class paradigms, DF-PEM integrates a PAE mechanism with the Mamba architecture to dynamically adapt to new categories while preserving vintage knowledge through memory-augmented representations. The SGS strategy further ensures robust prototype generation by modeling complex feature distributions and filtering noise, enabling stable knowledge retention. Additionally, the PRC adaptively balances prediction biases between old and new categories, enhancing generalization capabilities. Extensive experiments validate that DF-PEM achieves state-of-the-art performance across diverse multimodal remote sensing datasets, demonstrating significant improvements in mitigating catastrophic forgetting. These advancements not only bridge the gap between theoretical models and practical deployment but also pave the way for scalable and adaptive remote sensing systems in dynamic real-world scenarios.

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